UML 1 - Big Mammas Pizza

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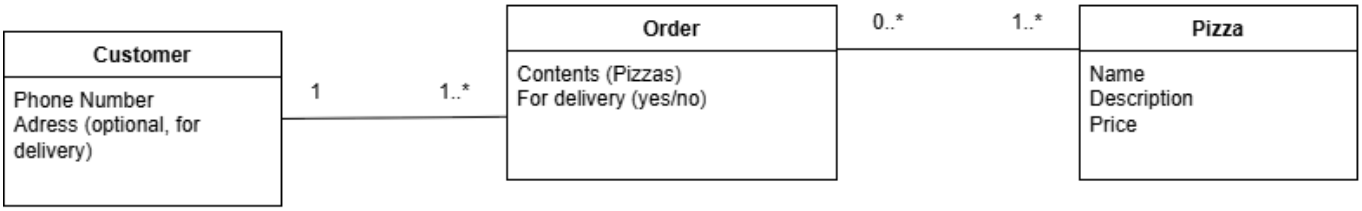
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1 User Stories

User Story	Acceptance criteria
As a customer, I would like to be able to order pizzas.	The system can create orders, that contain specified pizzas.
As a customer, I would like to be able to see a menu of available pizzas.	The system can output a list of pizzas
As a customer, I would like to be able to add extra toppings to ordered pizzas.	The system can append extra toppings to pizzas.
As a customer, I would like to be able to see the price of a pizza.	The system outputs price of a given pizza.
As a customer, I would like to be able to choose between delivery and pickup.	The system marks orders as delivery or pickup.
As a customer, I would like to be able to get an itemized receipt for an order.	Given an order, the system can show its contents and price per item.
As a customer, I would like to be able to add pizzas to an order	Orders can have pizzas added.
As a customer, I would like to see the total price of an order	The system can show the total price of an order.
As a customer, I would like to be able to remove pizzas from an order	Orders can have pizzas removed.
As a pizza baker, I would like to be able to see all orders	An order repository can show all orders

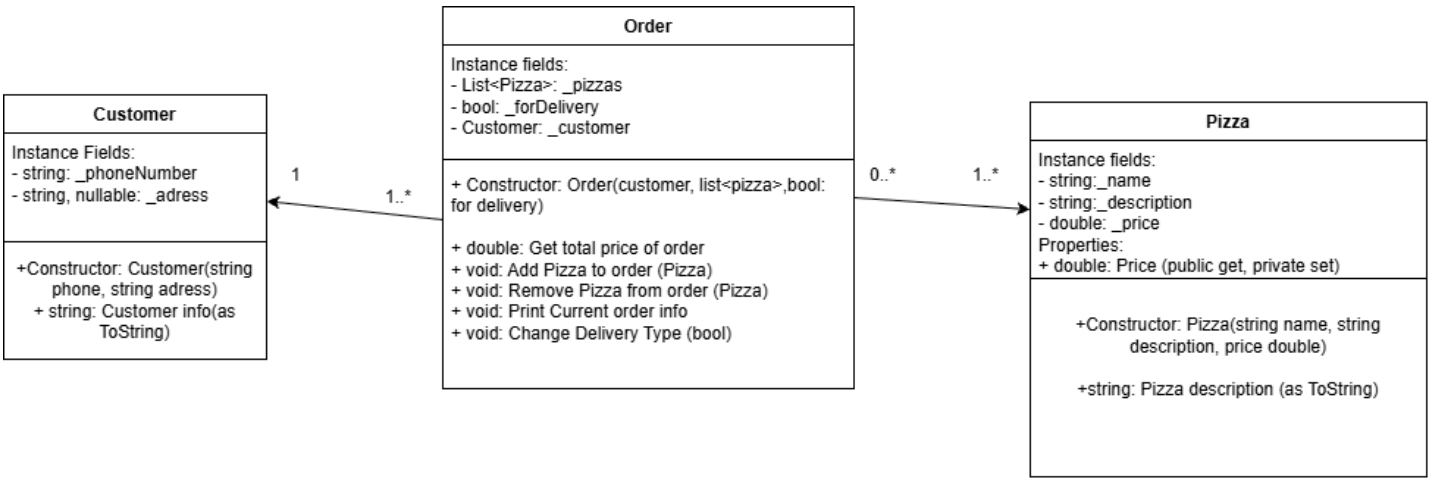
2 Domain Model

Simple domain model for the Pizza Store system:



3 Class Design Diagram

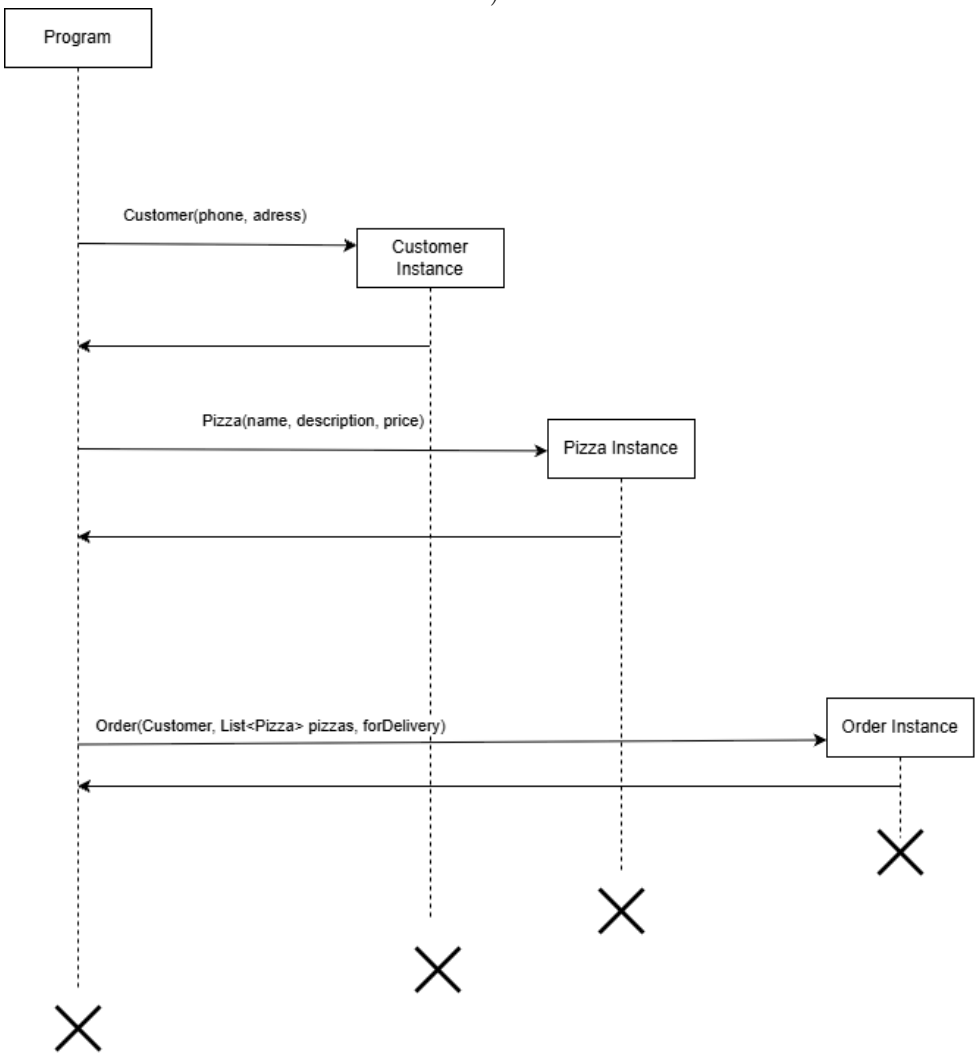
Class design diagram for the pizza store system:



It was chosen to have the order class hold a reference to the customer class and let the customer class be a small simple data container for customer information.
The only property is the public ***price*** property of the Pizza Class, as all other instance fields are interacted with through methods that access multiple fields at a time - there was no need for any other properties.

4 Interaction Diagram

Design sequence diagram showing the creation of a customer, pizza and order object instances.
The program simply creates the customer and pizza instances, and pass them to the constructor of an order object instance. As the order can hold multiple pizzas, a list object holding pizzas are created to contain the pizza object (not shown).



5 UML1 extended

5.1 1A

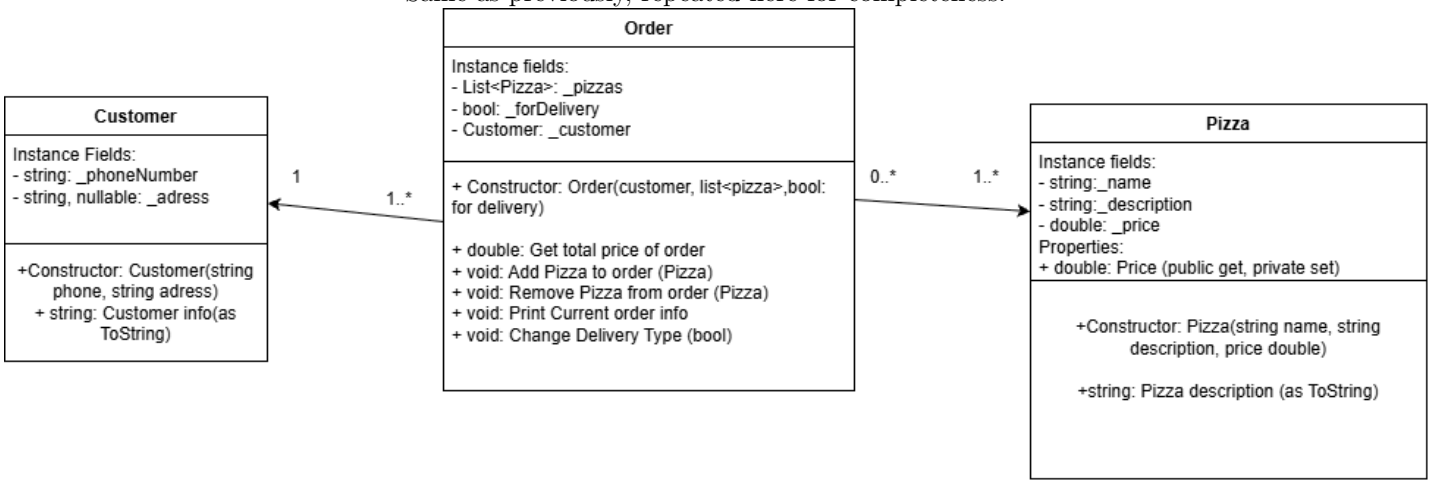
Implemented in UML1_26

5.2 1B

Implemented in UML1_26

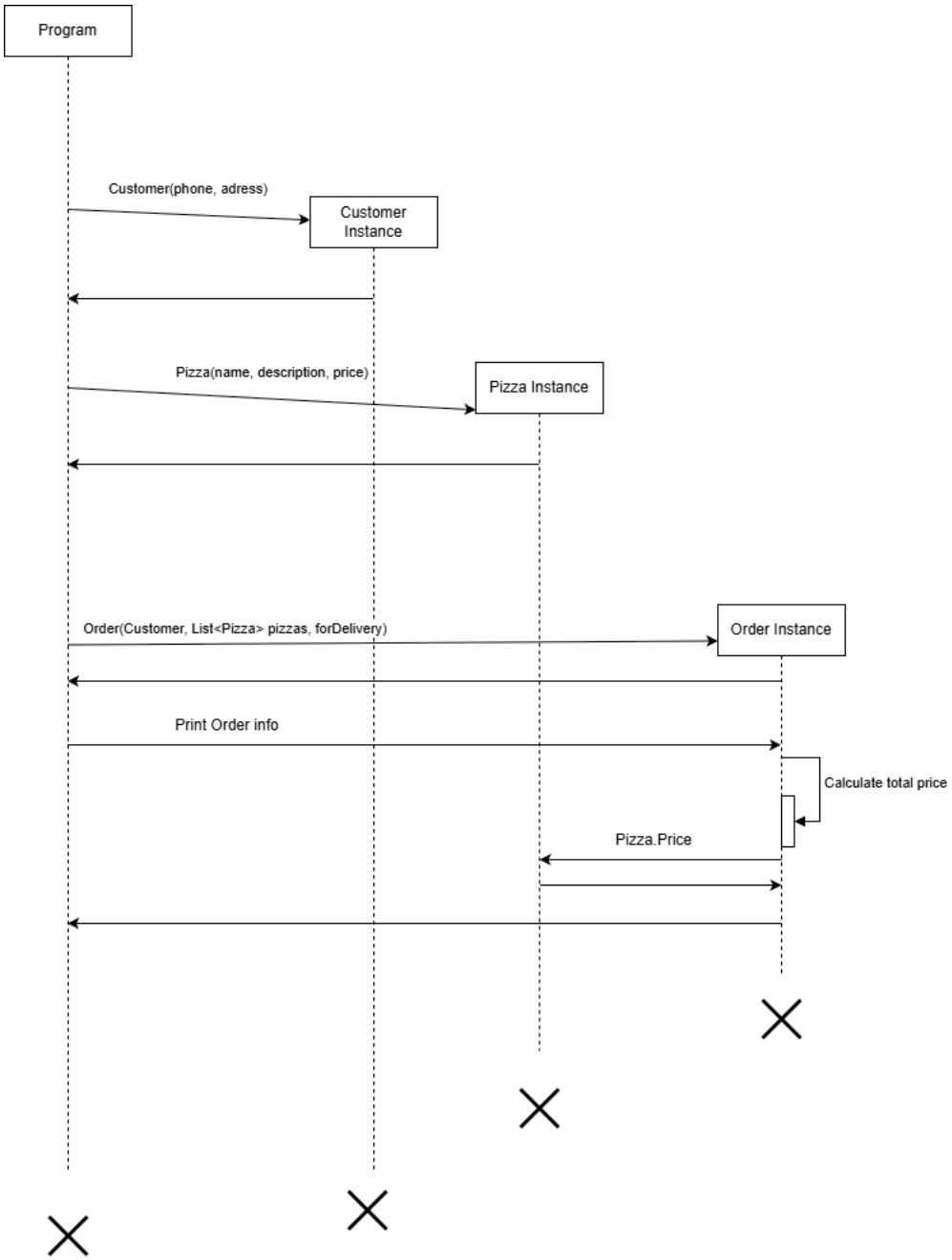
5.3 1C

Same as previously, repeated here for completeness:



5.4 1D

Design sequence diagram showing the creation of the customer, pizza and order objects, as well as the flow of control when calling the order instance for its Total price (next page):


Program
Customer(phone, adress)
Customer Instance
Pizza(name, description, price)
Pizza Instance
Order(Customer, List<Pizza> pizzas, forDelivery)
Order Instance
Print Order info
Calculate total price
Pizza.Price

5.5 2A

Implemented using a list in UML1_26_opg2.
To calculate the total price for an order that can contain multiple pizzas, a foreach loop over the pizza instances within an order are used.

5.6 3A

Implemented in UML1_26_opg3.
An orderline and topping class was implemented.
For comments, a comment instance field was added to the pizza class.

5.7 4A

Implemented in UML1_26_opg4.
Using a static private ***idcounter*** inside the Order class, orders are ensured unique IDs.

5.8 4B

Implemented in UML1_26_opg4.